Notes — Chemistry Chapter 6: Stoichiometry

Quick reference. Clean. No fluff. For the "why" behind each idea, see explanations.md.

Topic 1 — Empirical vs Molecular Formula

Definitions

- **Empirical formula:** the simplest whole-number ratio of atoms in a compound.
- **Molecular formula:** the actual number of each type of atom in one molecule of a compound.

Trick

To get empirical from molecular → divide all subscripts by their highest common factor (HCF).

Side-by-side

Compound	Molecular formula	Empirical formula
Water	H_2O	H_2O
Hydrogen peroxide	H_2O_2	HO
Glucose	$C_6H_{12}O_6$	CH_2O
Benzene	C_6H_6	CH
Ethene	C_2H_4	CH_2
Methane	CH_4	CH_4
Ammonia	NH_3	NH_3

Key point: If a molecular formula is already in lowest terms (HCF = 1), then molecular = empirical.

Topic 2 — Molecular Mass & Formula Mass

Definitions

- **Molecular mass:** sum of atomic masses of all atoms in one molecule. Unit = amu (atomic mass units).
- **Formula mass:** sum of atomic masses of all atoms in one formula unit of an ionic compound.
- They're the same calculation — just different names for different compound types.

How to calculate

1. Write the chemical formula.
2. Multiply each element's atomic mass by its subscript.
3. Add them all up.

Worked table

Compound	Type	Calculation	Mass
H_2O	molecular	2(1) + 16	18 amu
CO_2	molecular	12 + 2(16)	44 amu
H_2SO_4	molecular	2(1) + 32 + 4(16)	98 amu
NH_3	molecular	14 + 3(1)	17 amu
$C_6H_{12}O_6$	molecular	6(12) + 12(1) + 6(16)	180 amu
NaCl	ionic	23 + 35.5	58.5 amu
$Mg(OH)_2$	ionic	24 + 2(16) + 2(1)	58 amu
$CaCO_3$	ionic	40 + 12 + 48	100 amu
$KClO_3$	ionic	39 + 35.5 + 48	122.5 amu
$CuSO_4 \cdot 5H_2O$	ionic+hydrate	63.5 + 32 + 64 + 5(18)	249.5 amu

Common atomic masses to memorize

Element	Symbol	Mass
Hydrogen	H	1
Carbon	C	12
Nitrogen	N	14
Oxygen	O	16
Sodium	Na	23
Magnesium	Mg	24
Sulfur	S	32
Chlorine	Cl	35.5
Potassium	K	39
Calcium	Ca	40
Copper	Cu	63.5

Topic 3 — The Mole & Avogadro's Number

The definition

1 mole = 6.022×10^{23} particles of a substance.

This big number is called **Avogadro's number** (symbol: N_A).

Counting units you already know

Unit	Amount
Pair	2
Dozen	12
Score	20
Century	100
Ream	500
Mole	**602,200,000,000,000,000,000,000** (602 sextillion)

What can 1 mole contain?

- 1 mole of carbon atoms = 6.022×10^{23} C atoms

- 1 mole of water molecules = 6.022×10^{23} H_2O molecules
- 1 mole of NaCl = 6.022×10^{23} formula units of NaCl
- 1 mole of electrons = 6.022×10^{23} electrons

The "particles" can be atoms, molecules, formula units, or ions. The mole doesn't care what you're counting — it just counts.

Topic 4 — Gram Atomic / Molecular / Formula Mass (Molar Mass)

The core rule

The atomic/molecular/formula mass expressed in grams is the mass of 1 mole.

So if water's molecular mass is 18 amu, then 1 mole of water weighs exactly 18 grams.

Three names, one idea

Term	Used for	Example
Gram atomic mass	Elements (atoms)	C = 12 g
Gram molecular mass	Molecular compounds	H_2O = 18 g
Gram formula mass	Ionic compounds	NaCl = 58.5 g

All three are just **molar mass** — the mass of one mole of that substance.

The three golden equations

To find	Use
Moles from mass	n = mass ÷ molar mass
Mass from moles	mass = moles × molar mass
Particles from moles	particles = moles × 6.022×10^{23}
Moles from particles	moles = particles ÷ 6.022×10^{23}

Quick lookup — molar mass cheat sheet

Substance	Molar mass
H (atom)	1 g
H_2 (molecule)	2 g
C	12 g
N_2	28 g
O_2	32 g
H_2O	18 g
CO_2	44 g
NaCl	58.5 g
$CaCO_3$	100 g
Glucose $C_6H_{12}O_6$	180 g

Topic 5 — Formulas of Binary Ionic Compounds

The 4 steps

1. Write the cation (metal, +) first, then the anion (non-metal, −), with their charges.
2. Balance charges — the positive total must equal the negative total.
3. Write the balancing numbers as subscripts.
4. Drop the charges.

Worked mini-examples

Cation	Anion	Balance	Formula
Na^+	Cl^-	1:1	$NaCl$
K^+	O^{2-}	2:1	K_2O
Mg^{2+}	Cl^-	1:2	$MgCl_2$
Ca^{2+}	O^{2-}	1:1	CaO
Al^{3+}	O^{2-}	2:3	Al_2O_3
Al^{3+}	Cl^-	1:3	$AlCl_3$
Mg^{2+}	N^{3-}	3:2	Mg_3N_2

Common cations and anions to know

Cations	Charge	Anions	Charge
Li^+, Na^+, K^+	+1	F^-, Cl^-, Br^-	−1
Mg^{2+}, Ca^{2+}, Cu^{2+}	+2	O^{2-}, S^{2-}	−2
Al^{3+}	+3	N^{3-}, P^{3-}	−3
NH_4^+	+1	NO_3^-, NO_2^-	−1
		SO_4^{2-}	−2
		PO_4^{3-}	−3

Topic 6 — Chemical Equations & Balancing

A chemical equation

A symbolic way to describe what happens in a reaction:

reactants → products

- Reactants on the left.
- Products on the right.
- Arrow shows the direction of change.
- State symbols in brackets: (s), (l), (g), (aq).

Rules for balancing

1. Count atoms of each element on both sides.
2. Adjust **coefficients** (the big numbers in front) to make them equal.
3. **NEVER change subscripts** — that would change the substance itself.
4. Balance one element at a time.
5. Leave O and H for the end (usually easiest).

Worked example — methane combustion

Start: `CH₄ + O₂ → CO₂ + H₂O`

- C: 1 = 1 √
- H: 4 ≠ 2 (put 2 in front of H_2O)
- `CH₄ + O₂ → CO₂ + 2H₂O`
- O: 2 ≠ 4 (put 2 in front of O_2)
- `CH₄ + 2O₂ → CO₂ + 2H₂O` √ all balanced

More balanced examples

Reaction	Balanced equation
Hydrogen + Oxygen → Water	$2H_2 + O_2 \rightarrow 2H_2O$
Sodium + Water → NaOH + Hydrogen	$2Na + 2H_2O \rightarrow 2NaOH + H_2$
Iron + Oxygen → Iron oxide	$4Fe + 3O_2 \rightarrow 2Fe_2O_3$
Ammonia decomposition	$2NH_3 \rightarrow N_2 + 3H_2$
Magnesium + Oxygen → Magnesium oxide	$2Mg + O_2 \rightarrow 2MgO$

Topic 7 — Ionic Equations

The steps

1. Write the balanced chemical (molecular) equation.
2. Split every soluble (aqueous) compound into its ions.
3. Cancel ions that appear unchanged on both sides — these are **spectator ions**.
4. What's left is the **net ionic equation**.

Worked example — neutralization

Step	Equation
Molecular	$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$
Full ionic	$H^+ + Cl^- + Na^+ + OH^- \rightarrow Na^+ + Cl^- + H_2O$
Cancel spectators	Cl^- and Na^+ gone
Net ionic	**$H^+ + OH^- \rightarrow H_2O$**

Second example — precipitation

Step	Equation
Molecular	$AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$
Full ionic	$Ag^+ + NO_3^- + Na^+ + Cl^- \rightarrow AgCl(s) + Na^+ + NO_3^-$
Net ionic	**$Ag^+ + Cl^- \rightarrow AgCl(s)$**

Rule: solids (s), liquids (l), and gases (g) are NEVER split into ions. Only aqueous (aq) things split.

Topic 8 — Molecular vs Structural Formula

Definitions

- **Molecular formula:** counts the atoms. Example: butane = C_4H_{10}
- **Structural formula:** shows how atoms are connected. Example: $CH_3–CH_2–CH_2–CH_3$

Conversion steps (structural → molecular)

1. Identify all element types in the structural formula.
2. Count each element's atoms.
3. Write symbols with subscripts.

Examples

Structural	Molecular
$CH_3–CH_2–OH$	C_2H_6O
$CH_3–CH_2–NH_2$	C_2H_7N
$CH_3–CO–CH_3$	C_3H_6O
$CH_3–CH_2–CH_2–CH_3$	C_4H_{10}

Must-Remember Key Points

- Empirical formula = simplest ratio. Molecular formula = real count.
- Molecular/formula mass = sum of atomic masses of all atoms.
- **1 mole = 6.022 × 10^{23} particles** (Avogadro's number).
- Molar mass in grams = molecular/formula mass in amu (numerically equal).
- `n = mass ÷ molar mass` is the most important formula in this chapter.
- Balanced equation: atoms in = atoms out. Change coefficients only, never subscripts.
- In ionic equations, spectator ions cancel out.